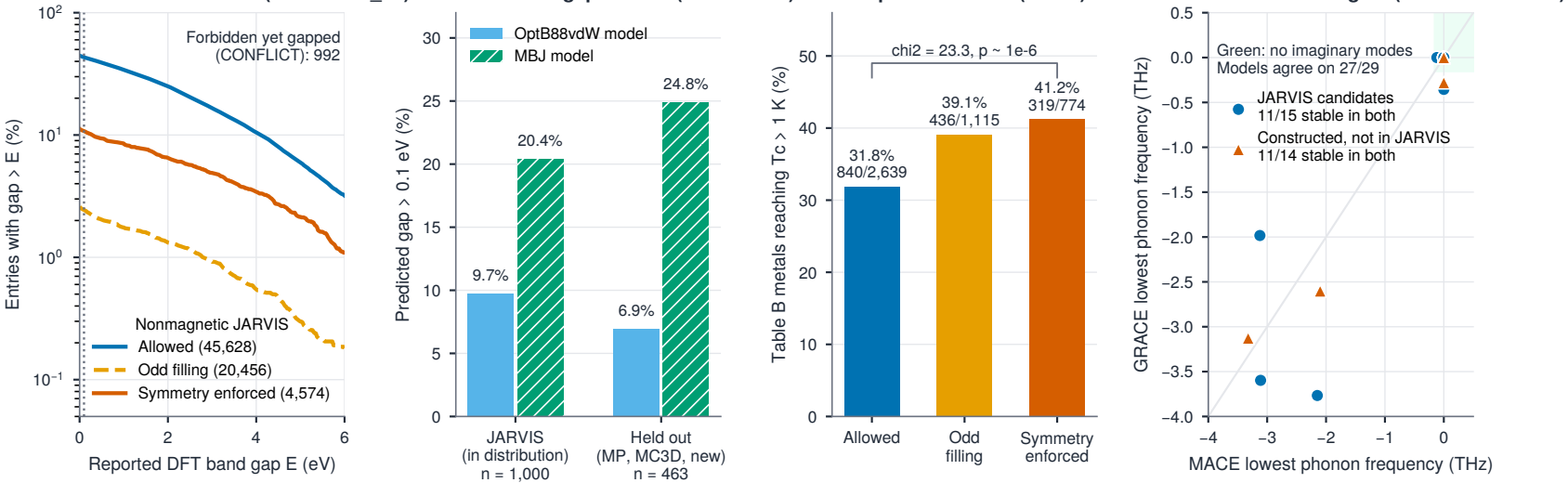
a Database census (JARVIS dft_3d)
Forbidden yet gapped
(CONFLICT): 992
Entries with gap > E (%)
Nonmagnetic JARVIS
Allowed (45,628)
Odd filling (20,456)
Symmetry enforced (4,574)
Reported DFT band gap E (eV)
b ML band gap models (ALIGNN 2.0)
OptB88vdW model
MBJ model
Predicted gap > 0.1 eV (%)
9.7%
20.4%
6.9%
24.8%
JARVIS
(in distribution)
n = 1,000
Held out
(MP, MC3D, new)
n = 463
c Superconductors (MC3D)
chi2 = 23.3, p ~ 1e-6
Table B metals reaching Tc > 1 K (%)
31.8%
840/2,639
39.1%
436/1,115
41.2%
319/774
Allowed
Odd
filling
Symmetry
enforced
d Two MLIPs agree (MACE vs GRACE)
Green: no imaginary modes
Models agree on 27/29
JARVIS candidates
11/15 stable in both
Constructed, not in JARVIS
11/14 stable in both
GRACE lowest phonon frequency (THz)
MACE lowest phonon frequency (THz)